

Technical Datasheet TnH Sensor Modules with Modbus RTU Protocol with RS485 Interface

The SN modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (this feature however also depends on the signal strength of the Modbus Master Device).

The RS485 SN-TnH module is sturdy, has low power usage and is easy to use.

TnH Sensor Module: -



The SN modules can be mounted in Outdoor or Indoor environment. **Soft setting** for **Modbus** is available in the software. In Addition, it has reset switch for Slave ID reset if forgotten.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power and Communication** port.

Specifications

General –

Dimensions	158mm L x 139 mm D
Power	Input Power – 12 - 24 V DC
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Configuration Settings: -

Communication Speed	Software Setting (9600 - 115200)
Data Bits	8
Parity	None
CRC	Yes
Slave ID	Software setting(1-247)

NOTE ON VPD UNITS:-

1. VPD (Vapor Pressure Deficit) Values are Measured in kPa (Kilopascals):
2. Conversion Clarification:
3. 1 kPa = 100 (sensor units).
4. Therefore, a sensor reading of 140 units corresponds to a VPD of 1.4 kPa.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40000	0X03	RS485	16 Bit Unsigned int
2	Humidity	40001	0X03	RS485	16 Bit Unsigned int
3	Display Baud Rate (Default:960)	40011	0x03	RS485	16 Bit Unsigned int
4	Enter New Baud Rate	40012	0x03	RS485	16 Bit Unsigned int
5	Display Slave ID (Default:1)	40013	0x03	RS485	16 Bit Unsigned int
6	Enter New Slave ID	40014	0x03	RS485	16 Bit Unsigned int
7	Display Parity (Default: None-3, Odd- 1, Even-2)	40015	0x03	RS485	16 Bit Unsigned int
8	Enter New Parity	40016	0x03	RS485	16 Bit Unsigned int
9	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16 Bit Unsigned int
10	Enter New Start/ Stop Bit	40018	0x03	RS485	16 Bit Unsigned int

Note: -

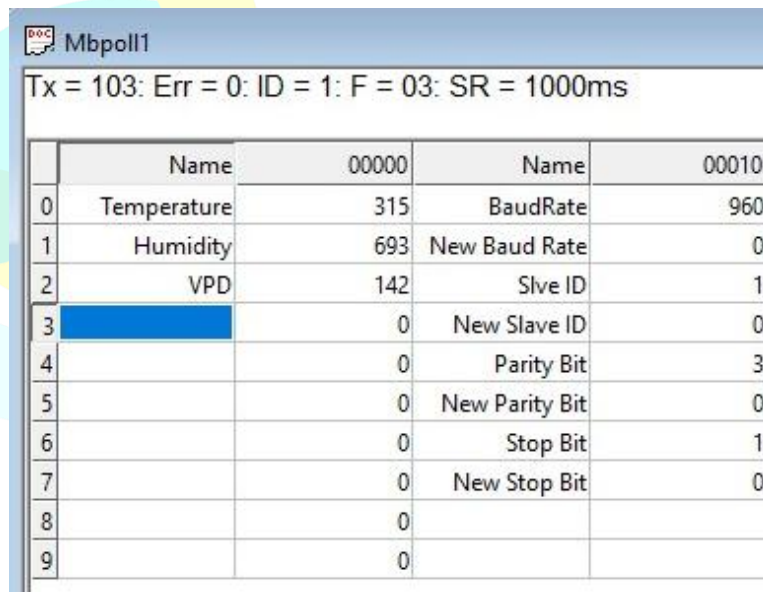
For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and Default slave id is 1).
- After connecting you will get data as shown below.



The screenshot shows the Mbpoll1 software window. At the top, it displays 'Tx = 103: Err = 0: ID = 1: F = 03: SR = 1000ms'. Below this is a table with 10 rows and 4 columns. The first two columns are 'Name' and '00000', and the next two are 'Name' and '00010'. The data is as follows:

	Name	00000	Name	00010
0	Temperature	315	BaudRate	960
1	Humidity	693	New Baud Rate	0
2	VPD	142	Slve ID	1
3		0	New Slave ID	0
4		0	Parity Bit	3
5		0	New Parity Bit	0
6		0	Stop Bit	1
7		0	New Stop Bit	0
8		0		
9		0		

- As you can see in the image above, resistor 40011 represents the present baud rate, resistor 40013 represents slave ID, resistor 40015 represents parity bit and resistor 40017 represents the Stop bit of the module.
- To change the baud rate and slave id you must enter a value on the 40012 resistor for the baud rate and 40014 for the slave id as shown in the image below
- Also, for changing the parity bit 40016 and stop bit 40018.

Write Single Register

Slave ID:

Address:

Value:

Result: N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers

Write Single Register

Slave ID:

Address:

Value:

Result: N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers

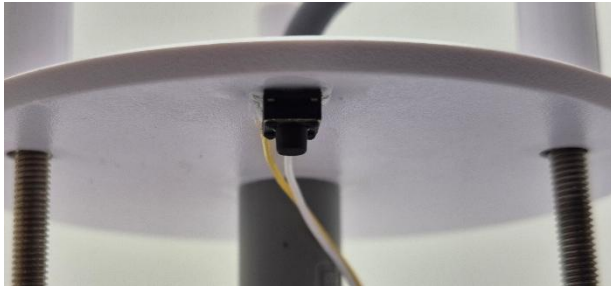
- After that press the reset button. Then the baud rate & slave ID will be changed.

Mbpoll1

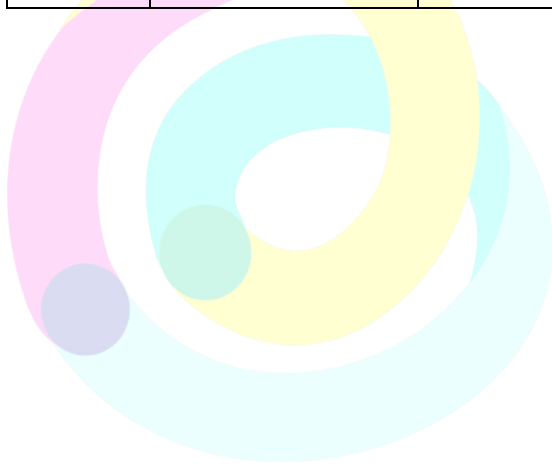
Tx = 4: Err = 0: ID = 24: F = 03: SR = 1000ms

	Name	00000	Name	00010
0	Temperature	315	BaudRate	11520
1	Humidity	684	New Baud Rate	0
2	VPD	146	Slve ID	24
3		0	New Slave ID	0
4		0	Parity Bit	3
5		0	New Parity Bit	0
6		0	Stop Bit	1
7		0	New Stop Bit	0
8		0		
9		0		

- Somehow you forgot the modbus settings of the board, in that case press reset button for 5 sec. Then you will get the default values for the system.



Sensor Name	Temperature Measuring Range	Humidity Measuring range	Operating Temperature	Measuring Accuracy
AHT20	-20°C ~ 85°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH
AHT21	-40°C ~ 120°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH



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